Simple Cayley permutations

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Abstract

We propose a notion of simplicity for Cayley permutations that is compatible with inflation and agrees, under a natural relational encoding, with indecomposability in the sense of Schmerl and Trotter. We establish a substitution decomposition analogous to that of permutations, in which the components are modules of the associated relational structure. We use it to enumerate simple Cayley permutations as well as simple restricted growth functions.

Keywords: Cayley permutation, simple permutation, inflation, substitution decomposition, relational structure

1 Introduction

A *Cayley permutation* is either empty or a word of positive integers that contains every number between one and its maximum value. In other words, a Cayley permutation is an endofunction ω on $[n] = \{1, 2, \ldots, n\}$ such that $\mathrm{im}(\omega) = [m]$ for some $m \leq n$. We write Cay_n for the set of Cayley permutations of length n and $\mathrm{Cay} = \bigcup_{n\geq 0} \mathrm{Cay}_n$. In addition, we write $\mathrm{Cay}_+ = \bigcup_{n\geq 1} \mathrm{Cay}_n$ for the set of nonempty Cayley permutations. There is a simple one-to-one correspondence between ballots (ordered set partitions) and Cayley permutations: a ballot ϖ corresponds to the Cayley permutation ω whose ith letter equals the unique index j such that i belongs to the jth block of ϖ. Cayley permutations may also be seen as representatives for equivalence classes of words modulo order isomorphism. For these and many other reasons, recent papers on Cayley permutations have explored their combinatorial and enumerative properties [3, 9, 10, 11, 12], as well as pattern avoidance [7, 8, 15].

The main objective of this paper is to define simple Cayley permutations and develop the surrounding theory. The notion of simplicity on (standard) permutations was introduced by Albert and Atkinson [1]: a permutation is *simple* if it has no proper interval of length at least two, where an interval is a set of consecutive positions whose corresponding values also form a contiguous set of integers. Every nonempty permutation of length n has trivial intervals of length 0 (the empty set), length 1 (single entries), and length n (the whole permutation). A permutation is simple if these are its only intervals. Simple permutations play an important role in the study of permutation classes (see the survey by Brignall [6] for a broad introduction). Albert

and Atkinson [1] showed that every permutation may be written as the inflation of a unique simple permutation. This allowed Albert, Atkinson, and Klazar [2] to enumerate simple permutations in terms of the compositional inverse of $f(x) = \sum_{n\geq 1} n!x^n$.

Here, we propose two natural substitution decompositions for Cayley permutations. One uses simple quotients and modules of an associated relational structure as parts; the other uses semisimple quotients and Hertzsprung intervals or singletons as parts. We begin with the latter decomposition, defining semisimplicity using Hertzsprung intervals in Section 2. In this decomposition, repeated quotient letters are necessarily inflated by the one-letter Cayley permutation. Inflation and the substitution decomposition are defined in Section 3. In Section 4, we derive a bivariate generating function identity for semisimple Cayley permutations and enumerate them. In Section 5, we view Cayley permutations as relational structures and show that indecomposable structures correspond to Cayley permutations that are simple. We also give the decomposition with a simple quotient, allowing constant components at repeated quotient letters. We then enumerate simple Cayley permutations. In Section 7, we show how inflation works for restricted growth functions and use this to enumerate the semisimple and simple ones.

The companion paper [14] is devoted to hereditary classes of Cayley permutations. There, we establish that a class of Cayley permutations with finitely many simple members has a finite basis and an algebraic generating function.

2 Hertzsprung intervals and semisimplicity

Throughout this paper, we will use the following terminology. Let ω be a word. We say that β is a *factor* of ω if there are words α and γ such that $\omega = \alpha\beta\gamma$. If α is empty we also say that β is a *prefix* of ω, and if, in addition, γ is nonempty then β is a *proper prefix* of ω. Similarly, if γ is empty we say that β is a *suffix* of ω, and that β is a *proper suffix* if, in addition, α is nonempty. The *standardization* of ω is the word $\mathrm{st}(\omega)$ obtained by replacing the smallest letter of ω by 1, the next smallest by 2, and so on, keeping repetitions equal. Note that $\mathrm{st}(\omega)$ is a Cayley permutation.

The second author [16] calls a permutation τ of length k a *Hertzsprung factor* of a permutation π if there is a nonnegative integer c and a factor $\beta = b_1 b_2 \cdots b_k$ of π such that $b_i - \tau(i) = c$ for each $i \in [k]$. That is, the factor β is a translate $\tau + c$ of τ.

We shall extend this definition to Cayley permutations. Let $\omega = w_1 \cdots w_n \in \mathrm{Cay}_n$. If $I = [a, b] \subseteq [n]$ is a nonempty positional interval, write

$$\omega|_I = w_a w_{a+1} \cdots w_b \qquad \text{and} \qquad \omega(I) = \{w_a, w_{a+1}, \ldots, w_b\}$$

for the factor of ω at positions I and for its set of values. More generally, if $S = \{i_1 < \cdots < i_r\} \subseteq [n]$, write $\omega|_S = w_{i_1} \cdots w_{i_r}$ for the corresponding subword and $\omega(S)$ for its values. We say that I is a *Hertzsprung interval*, or *H-interval*, of ω if there is a value interval $J = [c, d]$ such that

$$\omega(I) = J \quad \text{and} \quad \omega^{-1}(J) = I. \tag{1}$$

Following Brignall's [6] convention for permutation intervals, we also regard the empty set as an H-interval of every Cayley permutation. Thus, a nonempty H-interval

occupies consecutive positions and uses every value in a consecutive range, which we sometimes call a *value band*. Additionally, it contains every copy in ω of every value in J due to the second equality, which we call the *saturation condition*. In particular, $[n]$ is always an H-interval. A singleton $[i,i] = \{i\}$ is an H-interval precisely when the value $\omega(i)$ occurs nowhere else in ω. These two types of singletons (unique/repeated) will play different roles in inflations (see Section 3). Note also that the saturation condition is vacuous for permutations, so the usual definition of permutation interval is obtained as a special case of the one just given. Unlike in a permutation, the same factor may appear more than once in a Cayley permutation. For example, 12 appears in 1212 at both $[1,2]$ and $[3,4]$. We therefore specify an occurrence by its positional interval.

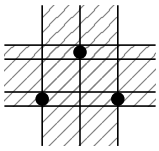
Let $\tau \in \mathrm{Cay}$. An *occurrence* of τ in ω is an H-interval I of ω where $\mathrm{st}(\omega|_I) = \tau$. Equivalently, for nonempty τ, we have

$$\omega|_I = \tau + c$$

for an integer $c \geq 0$ and no entry of ω outside I takes a value in $[c+1, c+\max(\tau)]$. We call τ a *Hertzsprung factor* of ω. Every H-interval is an occurrence of exactly one pattern, namely $\mathrm{st}(\omega|_I)$, and a singleton H-interval is an occurrence of 1. The empty pattern has the single occurrence $\emptyset$ in every word.

Example 2.1. In $\omega = 121$ the factor $\omega|_{[1,2]} = 12$ uses consecutive positions and consecutive values, but the value 1 occurs again in the third position. Hence the saturation condition fails, and $[1,2]$ is not an occurrence of 12. Similarly, $[1,2]$ is not an occurrence of 11 in 111. The nonempty H-intervals of 121 are the singleton $\{2\}$ and the whole word $[1,3]$, while the only nonempty H-interval of 111 is $[1,3]$.

The reader familiar with mesh patterns [5] and Cayley-mesh patterns [7] may notice that an occurrence of τ is an occurrence of the Cayley-mesh pattern whose underlying Cayley permutation is τ and where every box is shaded except for the four corner boxes. For instance, an occurrence of 121 is an occurrence of the Cayley-mesh pattern



and the algebraic condition (1) is a convenient definition of containment in this sense.

Let us briefly motivate the terminology just introduced. For permutations, a block in the sense of Albert and Atkinson [1], also commonly called an interval [6], occupies consecutive positions and carries a consecutive set of values, and is therefore an H-interval. We avoid both of the established names. "Block" is kept for the blocks of a ballot (where a nonempty H-interval is a union of consecutive blocks, see Proposition 2.3). "Interval" is used for positional and value intervals. Interval is also the name under which the permutation patterns literature treats these sets as the modules of a relational structure. This coincidence is special to permutations and does not extend to Cayley permutations, where the modules form a larger family than the H-intervals (see Section 5).

Definition 2.2. A Cayley permutation $\omega \in \mathrm{Cay}_n$ is *semisimple* if every H-interval of ω is empty, a singleton, or $[n]$; these H-intervals are called *trivial*. The word is

primitive [15] if adjacent letters are distinct. It is *simple* if it is semisimple and primitive, or if $\omega = 11$.

The empty word is semisimple and primitive, hence simple.

As noted before, for permutations the saturation condition is always satisfied and both semisimplicity and simplicity restrict to the usual definition of simplicity. The convention that 11 is simple makes simplicity equivalent to relational indecomposability at every length, as shown in Theorem 5.2. All Cayley permutations of length one and two are semisimple by definition. Example 2.1 shows that 121 and 111 are semisimple. An example of a non-semisimple Cayley permutation is 122: it contains the proper H-interval $[2, 3]$, which is an occurrence of 11 and occupies value band $[2, 2] = \{2\}$. 121 and 212 are simple, whereas 111 is semisimple but not simple. There are three semisimple Cayley permutations of length three and fifteen of length four, listed below:

$$111, 212, 121;$$

$$1111, 2212, 1121, 2121, 2122, 3132, 2131, 2132, 3142, 1211,$$
$$1212, 1312, 2312, 2313, 2413.$$

Among them, the simple ones are

$$212, 121;$$

$$2121, 3132, 2131, 2132, 3142, 1212, 1312, 2312, 2313, 2413.$$

Recall from Section 1 that Cayley permutations encode ballots. Namely, to the Cayley permutation $\omega \in \mathrm{Cay}_n$ we associate the ballot of $[n]$ defined by

$$\varpi = F_1 F_2 \cdots F_m, \qquad F_j = \omega^{-1}(j).$$

In this context, the fibres, F_j, are called *blocks*. For instance, the Cayley permutation $\omega = 211434125$ is associated with the ballot

$$\varpi = \{2, 3, 7\}\ \{1, 8\}\ \{5\}\ \{4, 6\}\ \{9\}.$$

We now interpret semisimplicity in terms of ballots, which will make the proofs of some results in the coming sections more convenient to state. To start, we characterize their H-intervals.

Proposition 2.3. *Let $\varpi = F_1 F_2 \cdots F_m$ be the ballot associated with $\omega \in \mathrm{Cay}_n$. Then a nonempty $I \subseteq [n]$ is an H-interval of ω if and only if I is an interval of $[n]$ and*

$$I = F_c \cup F_{c+1} \cup \cdots \cup F_d$$

for some $1 \leq c \leq d \leq m$. Furthermore, such an I is an occurrence of $\tau \in \mathrm{Cay}$ if and only if subtracting $\min(I) - 1$ from every element of the factor $F_c \cdots F_d$ turns it into the ballot of τ.

Proof. Let I be an H-interval, with value band $J = [c, d]$ as in (1). Then the saturation condition gives

$$I = \omega^{-1}(J) = \bigcup_{j=c}^{d} \omega^{-1}(j) = F_c \cup F_{c+1} \cup \cdots \cup F_d,$$

as wanted. Conversely, suppose that $I = F_c \cup \cdots \cup F_d$ is an interval of $[n]$, and put $J = [c, d]$. Every block is nonempty, so $\omega(I) = J$, and $\omega^{-1}(J) = F_c \cup \cdots \cup F_d = I$. Hence (1) holds and I is an H-interval.

For the second claim, let $a = \min(I)$. Position i of $\omega|_I$ corresponds to position $a+i-1$ of ω. Since $\omega(I) = [c, d]$, the letter j occurs in $\mathrm{st}(\omega|_I)$ at position i if and only if $\omega(a + i - 1) = c + j - 1$. Thus, the jth block of the ballot of $\mathrm{st}(\omega|_I)$ is obtained from F_{c+j-1} by subtracting $a - 1$ from each element. Therefore, subtracting $a - 1$ from every element of $F_c \cdots F_d$ gives the ballot of $\mathrm{st}(\omega|_I)$. This is the ballot of τ if and only if I is an occurrence of τ. □

In ballot terms, a nonempty H-interval is a run of consecutive blocks whose union is an interval of $[n]$. More precisely, a nonempty H-interval I covering the value band $\omega(I) = [c, d]$ corresponds to the blocks $F_c \cup F_{c+1} \cup \cdots \cup F_d$, whose union is I. Because each block is included in full, saturation is automatic. In particular, a singleton position is an H-interval precisely when it is a singleton block. In the example given before Proposition 2.3, the H-intervals of the Cayley permutation $\omega = 211434125$ correspond to those of the ballot $\varpi = \{2, 3, 7\}\{1, 8\}\{5\}\{4, 6\}\{9\}$ as follows:

$$\begin{aligned}
[1, 9] &\longleftrightarrow \varpi;\\
[1, 8] &\longleftrightarrow \{2, 3, 7\}\{1, 8\}\{5\}\{4, 6\};\\
[4, 6] &\longleftrightarrow \{5\}\{4, 6\};\\
[5, 5] &\longleftrightarrow \{5\};\\
[9, 9] &\longleftrightarrow \{9\}.
\end{aligned}$$

Thus, semisimplicity for ballots can be stated as follows.

Corollary 2.4. *Let $\omega \in \mathrm{Cay}_n$ and let ϖ be the corresponding ballot. Then ω is semisimple if and only if the only unions of consecutive blocks of ϖ that are intervals of $[n]$ are the empty set, the singleton blocks and $[n]$ itself.*

For instance, 121 has ballot $\{1, 3\}\{2\}$. The union $\{1, 3\}$ is not an interval of $[3]$, so the only nonempty H-intervals are the singleton block $\{2\}$ and $[3]$, and 121 is semisimple. More generally, in a semisimple Cayley permutation, no block of size two or more may be an interval of $[n]$ unless it is all of $[n]$.

3 Inflation and the substitution decomposition

Given a permutation σ of length k and nonempty permutations $\alpha_1, \ldots, \alpha_k$, the inflation of σ by $\alpha_1, \ldots, \alpha_k$ is the permutation $\sigma[\alpha_1, \ldots, \alpha_k]$ obtained by replacing each entry $\sigma(i)$ of σ with an interval that is order isomorphic to α_i, keeping the relative order of the $\sigma(i)$'s.

We shall define the inflation of Cayley permutations similarly, but with one additional constraint: only letters appearing exactly once in the Cayley permutation may be inflated by nonconstant words. This prevents positions carrying the same value from being expanded into competing value bands, which would make the process

ambiguous. Formally, given $\sigma \in \mathrm{Cay}_k$ we associate a nonempty Cayley permutation α_i with each position i, subject to the restriction

$$|\sigma^{-1}(\sigma(i))| > 1 \quad \Longrightarrow \quad \alpha_i = 1^{r_i} \text{ for some } r_i \geq 1. \tag{2}$$

The *inflation* $\sigma[\alpha_1, \ldots, \alpha_k]$ is the Cayley permutation obtained by replacing each entry $\sigma(i)$ with a factor order isomorphic to α_i, while preserving the relative order of the entries of σ. A singleton quotient fibre is replaced by a consecutive value band; a repeated fibre is replaced by a single value. We call the positional interval occupied by the entries corresponding to α_i the ith *part* of the inflation.

For the decomposition into H-intervals we impose the stronger restriction

$$|\sigma^{-1}(\sigma(i))| > 1 \quad \Longrightarrow \quad \alpha_i = 1. \tag{3}$$

Under this restriction every nonsingleton part is an H-interval. Throughout the remainder of this section and Section 4, inflations satisfy (3). For example,

$$312143[1, 1, 221, 1, 12, 1] = 4\,1\,332\,1\,56\,4.$$

The two copies of 1 and the two copies of 3 are forced to remain single letters due to (3); the unique copy of 2 is inflated by 221 and becomes the factor 332; and the unique copy of 4 is inflated by 12 to the factor 56. The six parts of the inflation are

$$\{1\}, \{2\}, [3, 5], \{6\}, [7, 8], \{9\}. \tag{4}$$

Now, let ω be a Cayley permutation. An *H-interval decomposition* of ω is a partition of its positions into consecutive intervals, each of which is an H-interval or a single position. Contracting the parts of an H-interval decomposition (and standardizing) yields a Cayley permutation, called the *quotient*. Repeated letters in the quotient come only from singleton parts. This distinction matches the two cases of (3): an H-interval part contracts to a letter that occurs once in the quotient, whereas a singleton part whose value occurs elsewhere contracts to a repeated letter. Contracting a Cayley permutation to obtain a quotient is the inverse of inflating the quotient. For instance, an H-interval decomposition of $\omega = 413321564$ is given by the six parts of (4). In general, the H-interval decomposition is not unique. The same ω admits the H-interval decomposition

$$4\,13321\,56\,4 \quad \longrightarrow \quad \{1\}, [2, 6], [7, 8], \{9\}$$

and contracting its parts yields the quotient 2132. Conversely, we see that

$$2132[1, 13321, 12, 1] = 4\,13321\,56\,4 = \omega.$$

Although H-interval decompositions need not be unique, requiring a semisimple quotient leaves only a controlled ambiguity: the quotient is unique, and the parts are unique except for the quotients 12 and 21, where an indecomposability condition restores uniqueness. This will be established below. We first record two elementary closure properties of H-intervals.

Lemma 3.1. *Let I and I' be two H-intervals of a Cayley permutation ω such that $I \cap I' \neq \emptyset$.*

1. *The union $I \cup I'$ is an H-interval.*

2. *The intersection $I \cap I'$ is an H-interval with $\omega(I \cap I') = \omega(I) \cap \omega(I')$.*

Proof. We use the ballot viewpoint of Proposition 2.3. Each H-interval of ω is a union of consecutive blocks of the ballot ϖ that form a set of consecutive elements. More precisely, write $\omega(I) = [c, d]$ and $\omega(I') = [c', d']$, so that I and I' are the runs of blocks $F_c \cdots F_d$ and $F_{c'} \cdots F_{d'}$. Suppose that I and I' overlap and let $\ell \in I \cap I'$. Then $F_c \cdots F_d$ and $F_{c'} \cdots F_{d'}$ share the block $F_{\omega(\ell)}$. Hence their union is again a union of consecutive blocks that form a set of consecutive elements, and Proposition 2.3 settles the first claim. Similarly, $I \cap I'$ is the run of blocks indexed by $[c, d] \cap [c', d']$, so Proposition 2.3 applies again and gives the second claim. □

Define direct and skew sums by

$$\alpha \oplus \beta = 12[\alpha, \beta] \quad \text{and} \quad \alpha \ominus \beta = 21[\alpha, \beta].$$

A Cayley permutation is *plus indecomposable* if it cannot be expressed as a direct sum of shorter Cayley permutations. Analogously, it is *minus indecomposable* if it cannot be expressed as a skew sum of shorter Cayley permutations.

The next two lemmas are used in the proof of Theorem 3.4.

Lemma 3.2. *Let $I = [1, b]$ be a nonempty proper H-interval of $\omega \in \mathrm{Cay}_n$, and suppose that its value band J contains 1 or $\max(\omega)$. Then $[b + 1, n]$ is an H-interval of ω. Furthermore, writing $\alpha = \mathrm{st}(\omega|_{[1,b]})$ and $\beta = \mathrm{st}(\omega|_{[b+1,n]})$, we have*

$$\omega = \begin{cases} \alpha \oplus \beta & \text{if } 1 \in J, \\ \alpha \ominus \beta & \text{if } \max(\omega) \in J. \end{cases}$$

Proof. Let $m = \max(\omega)$ and write the ballot of ω as $\varpi = F_1 F_2 \cdots F_m$. Write $J = [c, d]$. By Proposition 2.3,

$$I = F_c \cup F_{c+1} \cup \cdots \cup F_d.$$

Since the blocks partition $[n]$ and $I = [1, b]$, we have

$$[b + 1, n] = \begin{cases} F_{d+1} \cup \cdots \cup F_m & \text{if } c = 1, \\ F_1 \cup \cdots \cup F_{c-1} & \text{if } d = m. \end{cases}$$

These are the only cases under the hypothesis, and they are mutually exclusive because I is proper. In either case, $[b + 1, n]$ is a positional interval that is the union of consecutive blocks, so it is an H-interval by Proposition 2.3.

If $c = 1$, every value in I is smaller than every value in $[b + 1, n]$, and therefore $\omega = \alpha \oplus \beta$. If $d = m$, every value in I is larger than every value in $[b + 1, n]$, and therefore $\omega = \alpha \ominus \beta$. □

Lemma 3.3. *Let $\omega = \sigma[\alpha_1, \ldots, \alpha_k]$ be an inflation with parts $C_1, \ldots, C_k$, and let $T \subseteq [k]$ be a nonempty positional interval. Let $C = \bigcup_{i \in T} C_i$. Then*

$$C \text{ is an H-interval of } \omega \iff T \text{ is an H-interval of } \sigma.$$

Proof. For each value j of σ, let J_j be the value band of ω corresponding to j. These bands partition $[\max(\omega)]$, with $\max J_j < \min J_{j+1}$ for $1 \le j < \max(\sigma)$, and

$$\omega(C_i) = J_{\sigma(i)}.$$

First suppose that C is an H-interval of ω. Let $a = \min \sigma(T)$ and $b = \max \sigma(T)$. Since $a, b \in \sigma(T)$, we have $J_a \cup J_b \subseteq \omega(C)$. As $\omega(C)$ is a value interval, $J_j \subseteq \omega(C)$ for every $a \le j \le b$. If $\sigma(i) \in [a, b]$, it follows that $\omega(C_i) \subseteq \omega(C)$. The saturation of C then gives $C_i \subseteq C$, and hence $i \in T$. The reverse implication $i \in T \Rightarrow \sigma(i) \in [a, b]$ follows from the definitions of a and b. Therefore,

$$\sigma^{-1}([a, b]) = T.$$

Since σ is a Cayley permutation, every value in $[a, b]$ occurs in σ and therefore, by this equality, in $\sigma(T)$. Thus $\sigma(T) = [a, b]$, and T is an H-interval of σ.

Conversely, suppose that T is an H-interval of σ, with $\sigma(T) = [a, b]$ and $\sigma^{-1}([a, b]) = T$. Let

$$J = J_a \cup J_{a+1} \cup \cdots \cup J_b.$$

Because the value bands $J_1, \ldots, J_{\max(\sigma)}$ cover $[\max(\omega)]$, the union $J = J_a \cup \cdots \cup J_b$ equals $[\min J_a, \max J_b]$ and is therefore a value interval. Moreover,

$$\omega(C) = \bigcup_{i \in T} \omega(C_i) = \bigcup_{i \in T} J_{\sigma(i)} = J.$$

Every position p of ω belongs to exactly one part C_i, and $\omega(p) \in J$ if and only if $\sigma(i) \in [a, b]$. Since $\sigma^{-1}([a, b]) = T$, this is equivalent to $i \in T$. Hence $\omega^{-1}(J) = C$. Finally, C is a positional interval because it is a union of consecutive parts. Thus C is an H-interval of ω. □

Theorem 3.4. *Let ω be a Cayley permutation of length at least two. Then there is a unique semisimple Cayley permutation σ of length $k \ge 2$ such that*

$$\omega = \sigma[\alpha_1, \ldots, \alpha_k],$$

for some nonempty Cayley permutations $\alpha_1, \ldots, \alpha_k$ satisfying (3). If the quotient $\sigma \notin \{12, 21\}$, then the components α_i are uniquely determined. If $\sigma = 12$, they are unique if we require α_1 to be plus indecomposable. If $\sigma = 21$, they are unique if we require α_1 to be minus indecomposable.

Proof. Assume $\omega \in \mathrm{Cay}_n$ with $n \ge 2$. We first prove existence. Among all inflations $\omega = \sigma[\alpha_1, \ldots, \alpha_k]$ with $k \ge 2$, choose one with k minimal. Such an inflation exists, since $\omega = \omega[1, \ldots, 1]$. If σ were not semisimple, it would have an H-interval T with $2 \le |T| < k$. By Lemma 3.3, the union of the parts indexed by T is an H-interval of ω. Merge these parts into one, standardizing the factor on their union. The merged value band shares no value with any remaining part, so its letter in the new quotient occurs only once; the restrictions at all other repeated quotient letters are preserved. The resulting inflation has $k - |T| + 1$ parts. Since $2 \le |T| < k$, this number lies between 2 and $k - 1$, contradicting the minimality of k. Hence σ is semisimple.

We now prove uniqueness. Suppose first that ω is plus decomposable. Thus, for some $b \in [n-1]$, we have $\omega(i) < \omega(j)$ for all positions $i \le b < j$. Consider any expression of

ω as an inflation of a semisimple quotient with at least two parts. If b is a boundary between parts, the parts on either side express the quotient as a direct sum. If b lies inside a part, that part has at least two positions, so its quotient letter occurs only once by (3). Its value band contains values from positions on both sides of b. Since the value bands of distinct quotient letters are disjoint, every earlier quotient letter is smaller and every later quotient letter is larger. Separating the quotient immediately before or after this letter, choosing the option that leaves both sides nonempty, expresses it as a direct sum. If the quotient had length at least three, one summand would be a proper H-interval of length at least two. Thus a semisimple quotient must be 12.

Choose the smallest $b \in [n-1]$ with this property. The standardized prefix is plus indecomposable, or else its direct-sum decomposition would give a smaller such value of b. For any other $b' > b$ with this property, the standardized prefix ending at b' is plus decomposable, with its first summand ending at b. Hence b gives the unique decomposition whose first component is plus indecomposable. Reversing the value comparisons proves that a minus decomposable Cayley permutation has semisimple quotient 21 and a unique decomposition whose first component is minus indecomposable.

It remains to consider ω that is neither plus nor minus decomposable. Its maximal nonempty proper H-intervals are pairwise disjoint. Indeed, if two distinct ones overlapped, their union would be $[n]$ by Lemma 3.1 and maximality. One would be a proper prefix and the other a proper suffix. Their value bands cover $[\max(\omega)]$, and neither contains both extreme values, since saturation would make the corresponding H-interval equal to $[n]$. The prefix must therefore contain 1 or $\max(\omega)$, so Lemma 3.2 gives a direct-sum or skew-sum decomposition, a contradiction.

These maximal H-intervals, together with the singleton sets formed by the remaining positions, therefore partition $[n]$. Consider any inflation of ω with a semisimple quotient of length at least two. Each nonsingleton part is a proper H-interval and is therefore contained in a maximal one; each singleton part is contained in the unique member of the partition that contains its position. Thus the parts refine this partition. If a maximal H-interval were the union of at least two parts, Lemma 3.3 would give a proper H-interval of length at least two in the quotient, contradicting its semisimplicity. Consequently, the parts are exactly the maximal H-intervals and the uncovered singleton positions. Their standardized factors and their quotient are uniquely determined. □

The remaining semisimple quotient of length two, $\sigma = 11$, causes no ambiguity. Its two positions carry the same value, so (3) forces both components to be 1; the only permitted inflation is the Cayley permutation $11 = 11[1, 1]$ itself.

4 Enumeration of semisimple Cayley permutations

We shall adapt the enumeration of simple permutations [2] to Cayley permutations. The main difference is that, under (3), a position carrying a repeated letter in the quotient can only be inflated by the one-letter Cayley permutation 1. Given $\omega \in \mathrm{Cay}$,

let

$$\mu(\omega) = \#\{j : |\omega^{-1}(j)| = 1\}$$

denote the number of singleton fibres of ω, and define the bivariate ordinary generating function

$$B(x,t) = \sum_{\omega \in \mathrm{Cay}_+} x^{|\omega|} t^{\mu(\omega)} = \sum_{n \geq 1} B_n(t) x^n, \qquad B_n(t) = \sum_{\omega \in \mathrm{Cay}_n} t^{\mu(\omega)}.$$

The ballot viewpoint is convenient to determine an exponential generating function for the polynomials $B_n(t)$.

Proposition 4.1. *The polynomials $B_n(t)$ are determined by*

$$1 + \sum_{n \geq 1} B_n(t) \frac{x^n}{n!} = \frac{1}{2 - e^x - (t-1)x}.$$

Proof. View a Cayley permutation as a ballot. A singleton block has exponential generating function tx, while a block of size at least two has exponential generating function $e^x - 1 - x$. Taking a sequence of such nonempty blocks gives

$$\frac{1}{1 - (tx + e^x - 1 - x)} = \frac{1}{2 - e^x - (t-1)x}.$$

□

The first few of these polynomials are listed in Table 1.

n	$B_n(t)$
1	t
2	$1 + 2t^2$
3	$1 + 6t + 6t^3$
4	$7 + 8t + 36t^2 + 24t^4$
5	$21 + 100t + 60t^2 + 240t^3 + 120t^5$
6	$141 + 372t + 1170t^2 + 480t^3 + 1800t^4 + 720t^6$
7	$743 + 3584t + 5166t^2 + 13440t^3 + 4200t^4 + 15120t^5 + 5040t^7$

Table 1: The polynomials $B_n(t) = \sum_{\omega \in \mathrm{Cay}_n} t^{\mu(\omega)}$ for $n \leq 7$.

Let $\left\{ {n \atop k} \right\}$ denote the Stirling number of the second kind, that is, the number of partitions of $[n]$ into k blocks. Let $\left\{ {n \atop k} \right\}_{\geq 2}$ denote the number of such partitions in which every block has size at least two.

Proposition 4.2. *For $n \geq 1$,*

$$B_n(t) = \sum_{k=1}^{n} k! \sum_{j=0}^{k} \binom{n}{j} \left\{ {n-j \atop k-j} \right\} (t-1)^j.$$

Moreover, for $0 \leq j \leq n$,

$$[t^j]\, B_n(t) = \binom{n}{j} \sum_{i \geq 0} (i+j)! \left\{ {n-j \atop i} \right\}_{\geq 2}.$$

Proof. Using

$$(e^x-1)^r = r!\sum_{m\geq 0}\left\{{m \atop r}\right\}\frac{x^m}{m!},$$

we can expand the generating function in Proposition 4.1 as

$$\begin{aligned}
1+\sum_{n\geq 1}B_n(t)\frac{x^n}{n!} &= \sum_{k\geq 0}\left(e^x-1+(t-1)x\right)^k \\
&= \sum_{k\geq 0}\sum_{j=0}^{k}\binom{k}{j}(t-1)^j x^j (e^x-1)^{k-j} \\
&= \sum_{k\geq 0}\sum_{j=0}^{k}\binom{k}{j}(t-1)^j x^j (k-j)!\sum_{m\geq 0}\left\{{m \atop k-j}\right\}\frac{x^m}{m!} \\
&= 1+\sum_{n\geq 1}\left(\sum_{k=1}^{n}k!\sum_{j=0}^{k}\binom{n}{j}\left\{{n-j \atop k-j}\right\}(t-1)^j\right)\frac{x^n}{n!}.
\end{aligned}$$

Comparing coefficients of $x^n/n!$ proves the first formula. For the second formula, use

$$(e^x-1-x)^i = i!\sum_{m\geq 0}\left\{{m \atop i}\right\}_{\geq 2}\frac{x^m}{m!}.$$

Extracting the coefficient of t^j from the generating function and expanding gives

$$\begin{aligned}
[t^j]\left(1+\sum_{n\geq 1}B_n(t)\frac{x^n}{n!}\right) &= [t^j]\sum_{r\geq 0}\left(tx+e^x-1-x\right)^r \\
&= x^j\sum_{i\geq 0}\binom{i+j}{j}(e^x-1-x)^i \\
&= x^j\sum_{i\geq 0}\binom{i+j}{j}i!\sum_{m\geq 0}\left\{{m \atop i}\right\}_{\geq 2}\frac{x^m}{m!} \\
&= \sum_{n\geq j}\left(\binom{n}{j}\sum_{i\geq 0}(i+j)!\left\{{n-j \atop i}\right\}_{\geq 2}\right)\frac{x^n}{n!}.
\end{aligned}$$

Comparing coefficients of $x^n/n!$ proves the second formula.

Alternatively, the latter formula follows by a direct combinatorial argument. A ballot of $[n]$ with exactly j singletons is obtained by choosing the j elements in those blocks, partitioning the remaining $n-j$ elements into i blocks of size at least two, and linearly ordering all $i+j$ blocks. For each i, these choices give $\binom{n}{j}(i+j)!\left\{{n-j \atop i}\right\}_{\geq 2}$ ballots, and summing over i gives the formula. □

To enumerate semisimple Cayley permutations and several of their subclasses at once, let Q be a property of nonempty Cayley permutations such that $Q(1)$ holds and, for every inflation satisfying (3),

$$Q(\sigma[\alpha_1,\ldots,\alpha_k]) \quad\Longleftrightarrow\quad Q(\sigma) \text{ and } Q(\alpha_i) \text{ for every } i\in[k]. \tag{5}$$

Define

$$F_Q(x,t) = \sum_{\substack{\omega \in \mathrm{Cay}_+ \\ Q(\omega)}} x^{|\omega|} t^{\mu(\omega)}, \qquad H_Q(x,y) = \sum_{\substack{\sigma \text{ semisimple}, \ |\sigma| \geq 3 \\ Q(\sigma)}} x^{|\sigma| - \mu(\sigma)} y^{\mu(\sigma)}.$$

The two variables in H_Q distinguish positions in non-singleton fibres, which can only be inflated trivially, from positions in singleton fibres, which may be inflated by arbitrary components.

Lemma 4.3. *For a property Q satisfying $Q(1)$ and (5), let χ_ω be 1 if $Q(\omega)$ holds and 0 otherwise, for each word ω. Then*

$$F_Q(x,t) = tx + \chi_{11} x^2 + \frac{(\chi_{12} + \chi_{21}) F_Q(x,t)^2}{1 + F_Q(x,t)} + H_Q(x, F_Q(x,t)).$$

Proof. Write $F = F_Q(x,t)$. We partition the words counted by F according to the unique semisimple quotient supplied by Theorem 3.4.

The word 1 has weight tx. For the quotient 11, the restriction in (3) forces both components to be 1, so this quotient contributes $\chi_{11} x^2$.

Now suppose that $Q(12)$ holds, and let I_+ be the generating function for nonempty plus indecomposable words satisfying Q. Every word counted by F is uniquely either plus indecomposable or a direct sum $12[\alpha, \beta]$ in which α is plus indecomposable. Moreover, by (5), both components satisfy Q. Since the weight is multiplicative under direct sums, this gives

$$F = I_+ + I_+ F = I_+(1+F) \quad \Longrightarrow \quad I_+ = \frac{F}{1+F}.$$

Hence the words whose quotient is 12 contribute

$$I_+ F = \frac{F^2}{1+F}.$$

If $Q(12)$ does not hold, (5) excludes every nontrivial direct sum. The same argument with skew sums and minus indecomposable words applies to the quotient 21. Thus the quotients 12 and 21 together contribute

$$(\chi_{12} + \chi_{21}) \frac{F^2}{1+F}.$$

It remains to consider a semisimple quotient σ of length at least three. If a value occurs more than once in σ, then (3) forces the component at each of its positions to be 1; each such position contributes x. A value that occurs once may instead be inflated by any word counted by F, and therefore contributes F. There are $|\sigma| - \mu(\sigma)$ positions of the first kind and $\mu(\sigma)$ of the second. Hence the inflations of σ contribute

$$x^{|\sigma| - \mu(\sigma)} F^{\mu(\sigma)}.$$

By (5), such inflations satisfy Q exactly when σ does. Summing over the eligible quotients gives $H_Q(x,F)$. Since F has zero constant term, the displayed contribution has degree at least $|\sigma|$, hence this substitution is well defined as a formal series. Adding the four contributions proves the formula. □

We now take Q to be the property that holds for every nonempty Cayley permutation. In this unrestricted case, $F_Q = B$ and $\chi_{11} = \chi_{12} = \chi_{21} = 1$. Write $\widetilde{S}_{\geq 3} = H_Q$; explicitly,

$$\widetilde{S}_{\geq 3}(x, y) = \sum_{\substack{\sigma \text{ semisimple} \\ |\sigma| \geq 3}} x^{|\sigma| - \mu(\sigma)} y^{\mu(\sigma)}.$$

Lemma 4.3 now specializes as follows.

Theorem 4.4. *The generating function for semisimple Cayley permutations satisfies*

$$B(x,t) = tx + x^2 + \frac{2B(x,t)^2}{1 + B(x,t)} + \widetilde{S}_{\geq 3}(x, B(x,t)).$$

The identity in Theorem 4.4 rewrites as

$$\widetilde{S}_{\geq 3}(x, B(x,t)) = B(x,t) - tx - x^2 - \frac{2B(x,t)^2}{1 + B(x,t)}.$$

In their enumeration of simple permutations, Albert, Atkinson, and Klazar [2] use the coefficients of the compositional inverse of $\sum_{n \geq 1} n! x^n$. These coefficients were first studied by Comtet [17, p. 171]. We use the following general inversion lemma.

Lemma 4.5. *Let $F(x,t) = tx + \sum_{k \geq 2} F_k(t) x^k$, where $F_k(t) \in \mathbb{Z}[t]$. There is a unique formal power series $V(x) \in \mathbb{Z}[[x]]$ such that $F(x, V(x)) = x$. Its coefficients $v_n = [x^n]V(x)$ are given by $v_0 = 1$ and, for $n \geq 1$, by*

$$v_n = -\sum_{k=2}^{n+1} [x^{n+1-k}] F_k(V(x)),$$

where the right-hand side depends only on $v_0, \ldots, v_{n-1}$.

Proof. Substituting $t = V(x)$ into the given expansion of F turns the required identity $F(x, V(x)) = x$ into

$$xV(x) + \sum_{k \geq 2} F_k(V(x)) x^k = x.$$

Dividing by x and rearranging gives

$$V(x) = 1 - \sum_{k \geq 2} F_k(V(x)) x^{k-1}.$$

Its constant term gives $v_0 = 1$. For $n \geq 1$, extracting the coefficient of x^n gives the stated recurrence; terms with $k > n+1$ make no contribution. For each $2 \leq k \leq n+1$, the coefficient $[x^{n+1-k}]F_k(V(x))$ depends only on $v_0, \ldots, v_{n+1-k}$, and $n+1-k \leq n-1$. The recurrence therefore determines v_n from the preceding coefficients. Starting with $v_0 = 1$ constructs a unique solution coefficient by coefficient, and all its coefficients are integers because every F_k has integer coefficients. □

Corollary 4.6. *Under the hypotheses of Lemma 4.3, let $\widetilde{S}_Q(x)$ count the nonempty semisimple Cayley permutations satisfying Q, and let $V_Q(x)$ be the unique series with $F_Q(x, V_Q(x)) = x$. Then*

$$\widetilde{S}_Q(x) = 2x - xV_Q(x) + \frac{(\chi_{12} + \chi_{21})x^3}{1 + x}.$$

Proof. Since $Q(1)$ holds, the coefficient of x in $F_Q(x,t)$ is t, so Lemma 4.5 supplies V_Q. Substituting $t = V_Q(x)$ in Lemma 4.3 gives

$$H_Q(x,x) = x - xV_Q(x) - \chi_{11}x^2 - \frac{(\chi_{12}+\chi_{21})x^2}{1+x}.$$

The series H_Q includes only semisimple words of length at least three. Every Cayley permutation of length one or two is semisimple, so, using $Q(1)$, we have

$$H_Q(x,x) = \widetilde{S}_Q(x) - x - (\chi_{11}+\chi_{12}+\chi_{21})x^2.$$

Combining the two identities proves the formula; the contributions from 11 cancel. □

In the unrestricted case, where Q holds for every nonempty Cayley permutation, write $U = V_Q$. Thus U is the unique series with

$$B(x, U(x)) = x.$$

Write $u_n = [x^n]U(x)$. By the recurrence in Lemma 4.5, for example, $u_1 = -B_2(u_0) = -B_2(1) = -3$. The first terms of $U(x)$ are

$$\begin{aligned} U(x) = 1 - 3x - x^2 - 17x^3 - 95x^4 - 871x^5 - 9189x^6 \\ - 112111x^7 - 1537157x^8 - 23315921x^9 + \cdots \end{aligned}$$

Corollary 4.7. *Let $\tilde{s}_n$ be the number of semisimple Cayley permutations of length n and let $\widetilde{S}(x) = \sum_{n\geq 1} \tilde{s}_n x^n$. Then*

$$\widetilde{S}(x) = 2x + \frac{2x^3}{1+x} - xU(x).$$

Proof. This is Corollary 4.6 with $V_Q = U$, since both 12 and 21 satisfy the unrestricted property. □

Comparing coefficients in Corollary 4.7 gives

$$\tilde{s}_n = -u_{n-1} + 2(-1)^{n-1}$$

for $n \geq 3$. Hence

$$(\tilde{s}_n)_{n\geq 0} = 1, 1, 3, 3, 15, 97, 869, 9191, 112109, 1537159, 23315919, \ldots \tag{6}$$

For reference, Table 2 shows these values, alongside the Fubini numbers $|\mathrm{Cay}_n|$ and the number of simple Cayley permutations s_n of Theorem 6.1 below.

5 Modules and simplicity

A *relational structure* is a set together with a collection of binary relations on it. Analogues of simplicity and the substitution decomposition may be defined for any relational structure (e.g., graphs and posets). Schmerl and Trotter [20] defined *indecomposable* relational structures and proved that a relational structure is indecomposable if and only if it contains no nontrivial *module* (sometimes called an

n	$\lvert\mathrm{Cay}_n\rvert$	$\tilde{s}_n$	s_n
0	1	1	1
1	1	1	1
2	3	3	3
3	13	3	2
4	75	15	10
5	541	97	70
6	4683	869	634
7	47293	9191	6742
8	545835	112109	82306
9	7087261	1537159	1126846
10	102247563	23315919	17050626

Table 2: Cayley permutations $|\mathrm{Cay}_n|$, semisimple Cayley permutations $\tilde{s}_n$, and simple Cayley permutations s_n.

interval in the literature). Here, a module is a subset M of the structure such that, for every relation R, every $i, j \in M$, and every $x \notin M$,

$$iRx \iff jRx \qquad \text{and} \qquad xRi \iff xRj.$$

In other words, every two elements of M are ordered with respect to elements not in M in exactly the same way with respect to all relations. A module is nontrivial if it has at least two elements and is a proper subset of the ground set. We also allow the empty structure, which is indecomposable since it has no nontrivial module. Albert and Atkinson [1] encoded a permutation π of $[n]$ as the poset on $[n]$ where $x \prec y$ if and only if $x < y$ and $\pi(x) < \pi(y)$, so that π is simple exactly when this poset is indecomposable in the relational structure sense.

The presence of repeated entries requires a slightly different approach. To each Cayley permutation, we will associate a relational structure so that indecomposable structures correspond to Cayley permutations that are simple. Given a Cayley permutation $\omega \in \mathrm{Cay}_n$, define the relational structure

$$R(\omega) = ([n], <_1, <_2)$$

on the set of positions $[n]$, carrying two binary relations: the position order $i <_1 j$ if $i < j$, and the value order $i <_2 j$ if $\omega(i) < \omega(j)$. Both relations are irreflexive; $<_1$ is a linear order, and $<_2$ is a strict weak order whose ties are the positions of equal value. The main difference compared to Albert and Atkinson [1] is that the two orders are kept apart, and not merged into a single relation.

The empty set is a module, since its defining condition is vacuous. The next lemma characterizes the modules of $R(\omega)$.

Lemma 5.1. *The modules of $R(\omega)$ are the H-intervals of ω together with its nonempty positional intervals with constant value, which in the ballot are the nonempty positional intervals contained in a single block.*

Proof. The empty case is immediate. Let M be a nonempty module. If a position $x \notin M$ lay between two positions of M, then $y <_1 x$ for some $y \in M$ and $x <_1 y'$ for some $y' \in M$, contradicting the module condition. Thus M is a positional interval. If ω is constant on M, then M is contained in a single block. Otherwise, the module condition for $<_2$, in both directions, forces each position outside M to lie in a block below every block intersecting M or above every such block. In particular, no position outside M lies in a block intersecting M, so M is a union of whole blocks. Those blocks are consecutive, since a block between two of them would be nonempty and its positions neither below nor above all of M. By Proposition 2.3, M is an H-interval. Conversely, an H-interval is a module because every block outside it lies wholly below or wholly above it. A positional interval contained in a single block is also a module. □

Under the encoding employed by Albert and Atkinson [1], where $<_1$ and $<_2$ are merged into one relation, the correspondence between the intervals of a permutation and the modules of the corresponding relational structure is only partial. For instance, the positions 2 and 4 in 1432 form a module of the poset, but not an interval. Keeping $<_1$ and $<_2$ separate restores the exact correspondence in our setting. Indeed, if ω is a permutation then the only positional intervals with constant value are the singletons, which are H-intervals. Thus, for permutations the modules of $R(\omega)$ are exactly the H-intervals of ω. In general, the correspondence between modules and H-intervals breaks on Cayley permutations: a repeated value makes each single position carrying that value a module that is not an H-interval.

Theorem 5.2. *A Cayley permutation ω is simple if and only if the relational structure $R(\omega)$ is indecomposable.*

Proof. For $n \leq 2$, every Cayley permutation is simple and there is no nontrivial module. Suppose henceforth that $n \geq 3$. Suppose ω is simple, and let M be a module with $2 \leq |M| < n$. By Lemma 5.1, M is either an H-interval or a constant positional interval. The first is impossible because every nonempty H-interval of a semisimple Cayley permutation is a singleton or $[n]$. The second is impossible because a constant interval of length at least two contains a flat step, which primitivity forbids. No such M exists, so $R(\omega)$ is indecomposable.

Conversely, let $R(\omega)$ be indecomposable. Every H-interval is a module, so every H-interval is empty, a singleton, or $[n]$, which is to say that ω is semisimple. If $\omega(i) = \omega(i+1)$ for some i, then $\{i, i+1\}$ is a constant positional interval and hence a module; it is not a singleton, and it is not all of $[n]$ because $n \geq 3$, contradicting indecomposability. Hence ω is primitive. □

Next we allow the full range of components in the definition of inflation, subject only to (2). Every part is then a module by Lemma 5.1. Conversely, contracting the parts of a partition into nonempty modules gives such an inflation. For example,

$$1121 = 121[11, 1, 1], \qquad 111 = 11[1, 11].$$

We first relate the two kinds of quotient.

Lemma 5.3. *Every nonconstant semisimple Cayley permutation is obtained uniquely from a primitive simple Cayley permutation by replacing positions in repeated fibres by nonempty constant runs and leaving positions in singleton fibres unchanged.*

Proof. Contract every maximal constant run of a nonconstant semisimple word ω. Each contraction preserves semisimplicity since the inverse image of an H-interval under the order-preserving contraction map is an H-interval with the same value band. Thus a proper H-interval of length at least two after contraction would give one before contraction. The resulting word σ is primitive and semisimple, hence simple. A run of length greater than one cannot contain the entire fibre of its value, since it would be a proper H-interval of length at least two in ω. Its value therefore appears in at least two runs and remains repeated in σ.

Conversely, expand only positions in repeated fibres of a primitive simple word σ. Saturation forces any nonempty H-interval of the resulting word to contain every maximal run it meets. Contracting those runs gives an H-interval of σ. If this is a singleton, its value is unique in σ and its position was not expanded, so the original H-interval is a singleton. Otherwise it is the whole word. Thus the expanded word is semisimple. Maximal runs determine the contraction and their lengths uniquely. □

Theorem 5.4. *Every Cayley permutation ω of length at least two has a unique simple quotient σ of length $k \geq 2$ such that*

$$\omega = \sigma[\alpha_1, \ldots, \alpha_k]$$

for nonempty components satisfying (2). If $\sigma \notin \{11, 12, 21\}$, the components are unique. For $\sigma = 11$ they are unique on requiring $\alpha_1 = 1$; for $\sigma = 12$ or 21 they are unique on requiring α_1 to be plus or minus indecomposable, respectively.

Proof. A constant word has quotient 11, with the stated normalization. No nonconstant quotient can produce a constant word, and an inflation of 11 is constant, so this case is separate.

For a nonconstant word, take its H-interval decomposition from Theorem 3.4, with semisimple quotient τ. The quotient is nonconstant. Apply Lemma 5.3 to τ, merging the one-letter components along its constant runs. This gives an inflation of its simple contraction σ.

Conversely, start with an inflation of a simple nonconstant quotient σ. Replace each position in a repeated fibre by as many copies as there are letters in its constant component, while leaving each singleton-fibre position unchanged. The resulting quotient τ is semisimple by Lemma 5.3, and the same word is its inflation satisfying (3). The uniqueness in Theorem 3.4 determines τ and its components, subject to the two sum normalizations. Contracting its maximal runs recovers σ and the constant components uniquely. The quotients 12 and 21 have no repeated fibres, so their normalizations agree with those in Theorem 3.4. □

6 Enumeration of simple Cayley permutations

We now enumerate the simple Cayley permutations, which by Theorem 5.2 are exactly the Cayley permutations whose associated structure is indecomposable. Under (3),

primitivity satisfies (5). Indeed, every component of a primitive inflation is primitive because it is a standardized factor. Its quotient is primitive too; otherwise, adjacent equal letters in the quotient (which have forced one-letter components) would give adjacent equal letters in the inflation. Conversely, if the quotient and components are primitive, there are no equal adjacent letters within a component, and adjacent components use disjoint value bands because their quotient letters differ.

Write

$$P(x,t) = \sum_{\substack{\omega \in \mathrm{Cay}_+ \\ \omega \text{ primitive}}} x^{|\omega|} t^{\mu(\omega)} = \sum_{n \geq 1} P_n(t) x^n$$

for the primitive analogue of $B(x,t)$ and let

$$S_{\geq 3}(x,y) = \sum_{\substack{\sigma \text{ simple} \\ |\sigma| \geq 3}} x^{|\sigma| - \mu(\sigma)} y^{\mu(\sigma)}.$$

Theorem 6.1. *The generating function for simple Cayley permutations satisfies*

$$S_{\geq 3}(x, P(x,t)) = P(x,t) - tx - \frac{2P(x,t)^2}{1 + P(x,t)}.$$

Let $W(x)$ be the unique formal power series with $P(x, W(x)) = x$. Then the ordinary generating function $S(x) = \sum_{n \geq 1} s_n x^n$ for simple Cayley permutations is

$$S(x) = 2x + x^2 + \frac{2x^3}{1+x} - xW(x).$$

Proof. Apply the results of Section 4 to the property of being primitive, which satisfies (5) by the argument above. Here $F_Q = P$, the word 11 is excluded, and both 12 and 21 are admitted. Lemma 4.3 gives the first identity, Lemma 4.5 supplies W, and Corollary 4.6, with $V_Q = W$, counts the primitive semisimple words. Adding x^2 for the simple word 11 gives the second identity. □

The additional term x^2 in S accounts for 11, the only simple Cayley permutation that is not primitive.

To compute the coefficients of S, we express P in terms of B. Define

$$\widehat{B}(x,t) = \sum_{\omega \in \mathrm{Cay}_+} x^{|\omega| - \mu(\omega)} t^{\mu(\omega)} \quad \text{and} \quad \widehat{P}(x,t) = \sum_{\substack{\omega \in \mathrm{Cay}_+ \\ \omega \text{ primitive}}} x^{|\omega| - \mu(\omega)} t^{\mu(\omega)}.$$

Here t marks positions in singleton fibres and x marks all other positions. Contracting each maximal constant run to a single letter gives a unique primitive Cayley permutation. Conversely, we recover the original word by expanding each position into its corresponding nonempty constant run. A position in a singleton fibre may expand into a run of length one, contributing t, or a run of length at least two, whose value is then repeated. Summing the weights of the latter runs gives $x^2/(1-x)$, so the possible expansions of that position have total weight

$$t + \frac{x^2}{1-x},$$

whereas the possible expansions of each position carrying a repeated value have total weight $x/(1-x)$. Consequently,

$$\widehat{B}(x,t) = \widehat{P}\left(\frac{x}{1-x},\, t + \frac{x^2}{1-x}\right).$$

Since $B(x,t) = \widehat{B}(x,tx)$ and $P(x,t) = \widehat{P}(x,tx)$, we have

$$B(x,t) = P\left(\frac{x}{1-x},\, t(1-x) + x\right),$$

and inverting this substitution gives

$$P(x,t) = B\left(\frac{x}{1+x},\, (1+x)t - x\right).$$

Proposition 4.2 determines the coefficient polynomials of B, so this identity determines those of P. Lemma 4.5 then computes the coefficients of W, and Theorem 6.1 gives those of S.

Corollary 6.2. *The semisimple and simple quotient series satisfy*

$$\widetilde{S}_{\geq 3}(x,y) = S_{\geq 3}\left(\frac{x}{1-x}, y\right) + \frac{x^3}{1-x}.$$

Consequently,

$$B(x,t) = tx + \frac{x^2}{1-x} + \frac{2B(x,t)^2}{1+B(x,t)} + S_{\geq 3}\left(\frac{x}{1-x}, B(x,t)\right).$$

Proof. In Lemma 5.3, each repeated-fibre position contributes $x/(1-x)$ and each singleton-fibre position contributes y. The constant semisimple words of length at least three contribute $x^3/(1-x)$. This proves the first identity. Substitute it into Theorem 4.4 to obtain the second. Equivalently, use Theorem 5.4: its quotient 11 accounts for all constant words of length at least two, and each repeated position of a longer simple quotient may carry any nonempty constant run. □

The numbers s_n begin

$$(s_n)_{n\geq 0} = 1, 1, 3, 2, 10, 70, 634, 6742, 82306, 1126846, 17050626, 282097790, \ldots \quad (7)$$

agreeing with the number of associated structures $R(\omega)$ that are indecomposable at every length. We have found neither the sequence (7) in the OEIS [19] nor a closed form more explicit than the generating-function identity in Theorem 6.1.

7 Restricted growth functions

A nonempty Cayley permutation $\omega = w_1 \cdots w_n$ is a *restricted growth function*, or RGF, if $w_1 = 1$ and $w_{i+1} \leq \max(w_1, \ldots, w_i) + 1$ for $1 \leq i < n$. The empty word is also an RGF. Equivalently, in the ballot of ω the blocks appear in increasing order of their minima. Thus, the bijection between Cayley permutations and ballots

restricts to a bijection between RGFs and set partitions. We inherit the definition of semisimplicity from Cayley permutations. The enumeration of semisimple RGFs requires only minor changes: the blocks are unordered, and no RGF is a nontrivial skew sum, since every nonempty RGF starts with 1.

There is also an intrinsic version of the relational structure of Section 5 for set partitions. If ρ is a set partition of $[n]$, let

$$R(\rho) = ([n], <, \sim_\rho),$$

where $<$ is the natural order and $i \sim_\rho j$ if i and j belong to the same block of ρ.

Proposition 7.1. *Let ω be an RGF and ρ its corresponding set partition. A nonempty positional interval is an H-interval of ω if and only if it is a union of blocks of ρ. Moreover, $R(\rho)$ and $R(\omega)$ have the same modules. Consequently, $R(\rho)$ is indecomposable if and only if ω is simple.*

Proof. Let I be a nonempty positional interval that is a union of blocks. If blocks A, B, C satisfy $A, C \subseteq I$ and $\min A < \min B < \min C$, then $\min B \in I$, hence $B \subseteq I$. The blocks in I are consecutive in their order by minima, so I is an H-interval by Proposition 2.3. The converse follows from the same proposition.

The natural order forces every nonempty module M of $R(\rho)$ to be a positional interval. If M meets at least two blocks, the relation $\sim_\rho$ forces every block meeting M to be contained in M, so the first assertion makes M an H-interval. Otherwise ω is constant on M. Conversely, both kinds of interval are modules of $R(\rho)$. By Lemma 5.1, these are exactly the nonempty modules of $R(\omega)$; the empty set is a module of both structures. The final assertion now follows from Theorem 5.2. □

Lemma 7.2. *Assuming $\sigma \in \mathrm{Cay}_k$ and $\alpha_1, \ldots, \alpha_k \in \mathrm{Cay}$ are nonempty, an inflation $\sigma[\alpha_1, \ldots, \alpha_k]$ is an RGF if and only if $\sigma, \alpha_1, \ldots, \alpha_k$ are all RGFs.*

Proof. A Cayley permutation is an RGF precisely when, for each value v below its maximum, the first occurrence of v precedes the first occurrence of $v+1$. In $\omega = \sigma[\alpha_1, \ldots, \alpha_k]$ the values split into consecutive bands indexed, in increasing order, by the values of σ. Thus the RGF condition splits into conditions within the bands and between consecutive bands. The band indexed by j is a single value if the fibre $\sigma^{-1}(j)$ is not a singleton, in which case the corresponding components are constant. If $\sigma^{-1}(j) = \{i\}$, the values in the band first occur in the order in which they first occur in α_i. The conditions within the bands therefore say precisely that every α_i is an RGF. Suppose they hold. For either kind of fibre, the largest value of the jth band first occurs in the part indexed by $\min \sigma^{-1}(j)$. Since every α_i begins with 1, the smallest value of the $(j+1)$st band first occurs at the first position of the part indexed by $\min \sigma^{-1}(j+1)$. The parts are disjoint intervals occurring in index order, so the conditions between consecutive bands say precisely that $\min \sigma^{-1}(j) < \min \sigma^{-1}(j+1)$ for every j, which is to say that σ is an RGF. □

Theorem 7.3. *Let $B_n^{\mathrm{RGF}}(t) = \sum_\omega t^{\mu(\omega)}$, the sum being over the RGFs ω of length n, let $B^{\mathrm{RGF}}(x,t) = \sum_{n\geq 1} B_n^{\mathrm{RGF}}(t) x^n$, and let $\widetilde{S}_{\geq 3}^{\mathrm{RGF}}$ and $\widetilde{S}^{\mathrm{RGF}}$ be defined as $\widetilde{S}_{\geq 3}$ and $\widetilde{S}$ with "semisimple" replaced by "semisimple RGF". Then*

$$1 + \sum_{n\geq 1} B_n^{\mathrm{RGF}}(t) \frac{x^n}{n!} = \exp(tx + e^x - 1 - x),$$

and, for $n \geq 1$,

$$B_n^{\mathrm{RGF}}(t) = \sum_{k=1}^{n} \sum_{j=0}^{k} \binom{n}{j} \left\{ {n-j \atop k-j} \right\} (t-1)^j.$$

Moreover, for $n \geq 1$ *and* $0 \leq j \leq n$,

$$[t^j]\, B_n^{\mathrm{RGF}}(t) = \binom{n}{j} \sum_{i \geq 0} \left\{ {n-j \atop i} \right\}_{\geq 2}.$$

Furthermore,

$$\widetilde{S}_{\geq 3}^{\mathrm{RGF}}(x, B^{\mathrm{RGF}}(x,t)) = B^{\mathrm{RGF}}(x,t) - tx - x^2 - \frac{B^{\mathrm{RGF}}(x,t)^2}{1 + B^{\mathrm{RGF}}(x,t)},$$

and, letting $U^{\mathrm{RGF}}(x)$ *be the unique formal power series with* $B^{\mathrm{RGF}}(x, U^{\mathrm{RGF}}(x)) = x$,

$$\widetilde{S}^{\mathrm{RGF}}(x) = 2x + \frac{x^3}{1+x} - xU^{\mathrm{RGF}}(x).$$

Proof. The blocks of a set partition are unordered, so the sequence of blocks in the proof of Proposition 4.1 is replaced by a set of blocks and $1/(1-u)$ becomes e^u; in the two formulas of Proposition 4.2 the same change removes the factors $k!$ and $(i+j)!$.

Lemma 7.2 verifies (5) for the RGF property, which holds for 1, 11, and 12, but not for 21. The remaining identities and the existence and uniqueness of U^{RGF} follow from Lemma 4.3, Lemma 4.5, and Corollary 4.6, with $F_Q = B^{\mathrm{RGF}}$ and $V_Q = U^{\mathrm{RGF}}$. □

Writing $u_n^{\mathrm{RGF}} = [x^n]U^{\mathrm{RGF}}(x)$, the coefficients $\tilde{s}_n^{\mathrm{RGF}} = [x^n]\widetilde{S}^{\mathrm{RGF}}(x)$, for $n \geq 1$, satisfy

$$\tilde{s}_n^{\mathrm{RGF}} = -u_{n-1}^{\mathrm{RGF}} + (-1)^{n-1}$$

for $n \geq 3$. Including $\tilde{s}_0^{\mathrm{RGF}} = 1$, the sequence begins

$$(\tilde{s}_n^{\mathrm{RGF}})_{n \geq 0} = 1, 1, 2, 2, 4, 13, 51, 228, 1129, 6093, 35351, 218467, \ldots$$

and it is not in the OEIS [19].

For simple RGFs, we use the same method as for the coefficients s_n in Theorem 6.1. Define

$$P^{\mathrm{RGF}}(x,t) = \sum_{\substack{\omega \in \mathrm{Cay}_+ \\ \omega \text{ primitive RGF}}} x^{|\omega|} t^{\mu(\omega)},$$

and let W^{RGF} satisfy $P^{\mathrm{RGF}}(x, W^{\mathrm{RGF}}(x)) = x$. Run expansion and Corollary 4.6, adding x^2 for 11, give

$$P^{\mathrm{RGF}}(x,t) = B^{\mathrm{RGF}}\left(\frac{x}{1+x}, (1+x)t - x \right),$$

$$S^{\mathrm{RGF}}(x) = 2x + x^2 + \frac{x^3}{1+x} - xW^{\mathrm{RGF}}(x).$$

Lemma 5.3 preserves the RGF property, so the first identity of Corollary 6.2 also holds with RGF superscripts. Its second identity holds with B^{RGF} and $S_{\geq 3}^{\mathrm{RGF}}$, and with coefficient 1 instead of 2 on the sum term, since no RGF is a nontrivial skew sum.

Thus, writing $s_n^{\mathrm{RGF}} = [x^n]S^{\mathrm{RGF}}(x)$ for $n \geq 1$ and including $s_0^{\mathrm{RGF}} = 1$, we obtain

$$(s_n^{\mathrm{RGF}})_{n\geq 0} = 1, 1, 2, 1, 1, 5, 16, 69, 316, 1591, 8614, \ldots$$

This sequence is related to OEIS A099947, the numbers t_n of topologically connected set partitions [19]. These partitions have no proper subinterval that is a union of blocks; Beissinger [4] calls them irreducible, and Dykema [18] calls them connected. By Proposition 7.1, for $n \geq 2$ these partitions correspond to semisimple RGFs with no singleton fibres. Hence their generating function is

$$T(x) = \sum_{n\geq 1} t_n x^n = x + x^2 + \widetilde{S}_{\geq 3}^{\mathrm{RGF}}(x, 0).$$

Combining run expansion with the deletion of marked singleton blocks gives

$$P^{\mathrm{RGF}}(x, t) = x + (1 + x)B^{\mathrm{RGF}}(x, (1 + x)(t - 1)).$$

Substituting $t = W^{\mathrm{RGF}}(x)$ in this identity and using Theorem 7.3 gives

$$T(x) = (1 + x)S^{\mathrm{RGF}}(x) - 2x^2 - 2x^3.$$

Thus $t_n = s_n^{\mathrm{RGF}} + s_{n-1}^{\mathrm{RGF}}$ for $n \geq 4$.

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